Sarthak Jain

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Summary:

Software Engineer with 9+ years of experience building large-scale distributed systems. Skilled in full-stack development with a strong focus on backend engineering, API development, performance optimization, and team leadership. Passionate about innovation, GenAI-driven tooling and operational efficiency.

Technical Skills:

Languages: Java, JavaScript, Shell Scripting, Python, MySQL, HTML, CSS
Technologies: IntelliJ, AWS, Git, CI/CD
Databases: MySQL Server, S3, DynamoDB
GenAI & ML: Claude CLI, Amazon Kiro, Word2Vec, Weblab A/B testing experimentation framework

Professional Experience:

Amazon.com Services Inc, New York, NY, USA — **June 2019 – Present**

Software Development Engineer II

Designed and developed large-scale, low-latency backend services, REST APIs, and frontend components that power delivery messaging for 200M+ global users on Amazon's retail websites, leveraging internal GenAI tools (Amazon Q/Kiro) to accelerate implementation. Led end-to-end full-stack feature delivery and A/B experiments while improving operational excellence through monitoring, production troubleshooting, high-availability optimizations, CI/CD automation, and clear documentation of key architectural decisions. [Java, JavaScript, Object Oriented Design, AWS]

Checkout Delivery Message Platform Team

- Led design and development of 1-hr and 3-hr Sub-Same Day delivery messaging CX on Amazon's Checkout Page.
- Spearheaded design and development of a unified mobile bottom-sheet experience that simplifies delivery scheduling across all fulfillment programs on the Checkout Page.
- Modernized Checkout delivery option CX by converting legacy radio buttons to container-style layouts and simplifying Sub-Same Day scheduling through a radio button experience, enhancing accessibility (a11y-compliace), usability and customer interaction rates.

Pre-Checkout Delivery Message Platform Team

- Launched delivery messaging on the Cart Page by designing and developing new APIs, achieving a **1.11% reduction in checkout time** and driving major business impact: **+$170M Gross Customer Contribution Profit (GCCP), +$377M Operating Profit Savings (OPS), +15M paid units, and 10K fewer customer contacts.**
- Led design, implementation and launch of multiple high-impact Weblab based experiments to enhance delivery messaging customer experience across Amazon's Retail Websites. [Java, JavaScript, Datapath]
  - Simplified delivery condition messaging **(+$77M GCCP, +$372M OPS, 141K fewer customer contacts resulting in 7K fewer concessions)**
  - Onboarded Subscribe and Save delivery messages to Unified Delivery Messaging framework **(+$23M Annualized GCCP and +1.6M WW Annualized SnS subscriptions)**
  - Onboarded Multi Offer Display collapsed delivery message to Unified Delivery Messaging framework **(+$22M Annualized GCCP)**
  - Integrated Prime badge and delivery messaging into Prime Exclusive Deals on Detail Page and Search Page
  - Added Prime Upsell CX for Non-Prime and Unrecognized customers on Detail Page and Search Page
- Designed and developed CX Replay Testing framework automating regression testing for all customer-facing changes. [Java, JavaScript, AWS - Lambda, EC2, ECS, Fargate, S3, DynamoDB]
- Drove Operational Excellence initiatives across SDX organization for 1.5 years, establishing Away Team metrics monitoring and reporting processes as well as prediction band metrics and automated alarms.
- Autonomously designed, built and delivered workflows for updating Prime badge and delivery messaging information when quantity is changed on Detail Page of Amazon's Retail Websites for any product. [Java, JavaScript, Perl]
- Mentored cross-functional peers (SDEs, TPMs, PMs) on design reviews, domain knowledge, debugging techniques and service ownership best practices.

Morgan Stanley Services Group, New York, NY, USA — **Aug 2016 – Apr 2019**

Technology Associate

- Developed backend REST API services for recommendation system (Next Best Action) providing investment options to Financial Advisors. [Java, Scala, Solr, Hive, MySQL, AngularJS]
  - Developed **Edge NGrams Typeahead and Search functionality** with phrase matching and **response times of 15ms**
  - Enabled near real time recommendation of options by transforming batch processing to Spark processing framework
  - Reduced Solr indexing process runtime by 50% for 140 million documents and optimized API query response times
  - Handled additional responsibility of Level 3 Support lead: managed a team of 3, coordinated with multiple teams and resolved any issues escalated by L2 across all components
- Developed POC to improve search relevancy using **Word2Vec ML algorithm**. Demonstrated Word2Vec query pipeline improved relevancy using AngularJS UI dashboard, hence **POC was productionized.**
- Transformed Spark SQL/Scala code to optimized Hive Script reducing the run time by 50%.
- Completed the coding intensive Technology Analyst program which included Java, Scala, C++, and SQL, etc.
- Developed feed monitoring dashboard using Java, Spring Framework, REST API, HTML, Vis.JS and AngularJS.

S&P Global Market Intelligence, New York, NY, USA **Jun 2015 – Dec 2015**

Credit Solutions Architecture Intern

- Worked in team of 3 using Agile methodology, designed & developed web application to visualize and analyze relational database in detailed graph format. Application was used to query on properties and relationships between different nodes and edges. Displayed the queried information in a time series format by varying the node sizes and colors.
  [Java, Servlets, Neo4j Graph Database, Cypher, MySQL, JavaScript, Vis.JS, AngularJS, HTML, CSS]

Personal Projects:

- **BookFinder - Book Recommendation Engine - Live Demo: https://sarthak-jain.github.io/BookFinderApplication/**
  Built a full-stack book recommendation engine on a Neo4j graph with 60K books, 129K nodes, and 376K relationships from the Goodreads dataset, featuring Lucene-powered **full-text search, filters, and interactive graph visualization**. Implemented four recommendation strategies (**graph traversal, shelf similarity, collaborative filtering, and weighted hybrid**) plus mood-based discovery (10 curated moods and custom mood builder).
  [Claude CLI, Java, Cypher, REST, deployed via React (GitHub Pages), Spring Boot (Railway), and Neo4j AuraDB]

Education:

- **New York University** **-** Master of Science in Computer Science Sep 2014 – May 2016
  Graduate School of Arts and Sciences **-** Courant Institute of Mathematical Sciences, New York, USA
  Course Work: Fundamental Algorithms, Realtime and Big Data Analytics, Advanced Databases, Production Quality Software, Natural Language Processing, Big Data Science
- **Manipal University -** Bachelor of Engineering in Information Technology July 2009 – May 2013
  Manipal Institute of Technology, Manipal, India
  Course Work: Data Structures, Object Oriented Programming, Data Mining, Distributed Systems